

OVERVIEW OF LECTURE #1



Supervised Learning

Training a computer using labeled data, where the outcome is already known.

Example-Utilizing historical patient records with known outcomes (labels) to predict health outcomes.



Unsupervised Learning

Allowing a computer to analyze data without explicit guidance, looking for patterns or structures.

Letting a computer sift through patient data to find clusters of similar symptoms without any prior labeling.



Reinforcement Learning

A computer learns through trial and error, receiving feedback for its actions to optimize outcomes over time.

A computer testing different public health strategies and adapting based on the success of each strategy, like improving vaccination campaigns based on feedback.

FUNCTION

In mathematics, a function is a relation between input x and output y sets, where each x maps to a unique y , denoted as $y = f(x)$.

This notation means that for a given 'x' value, the function 'f' will produce a 'y' value.